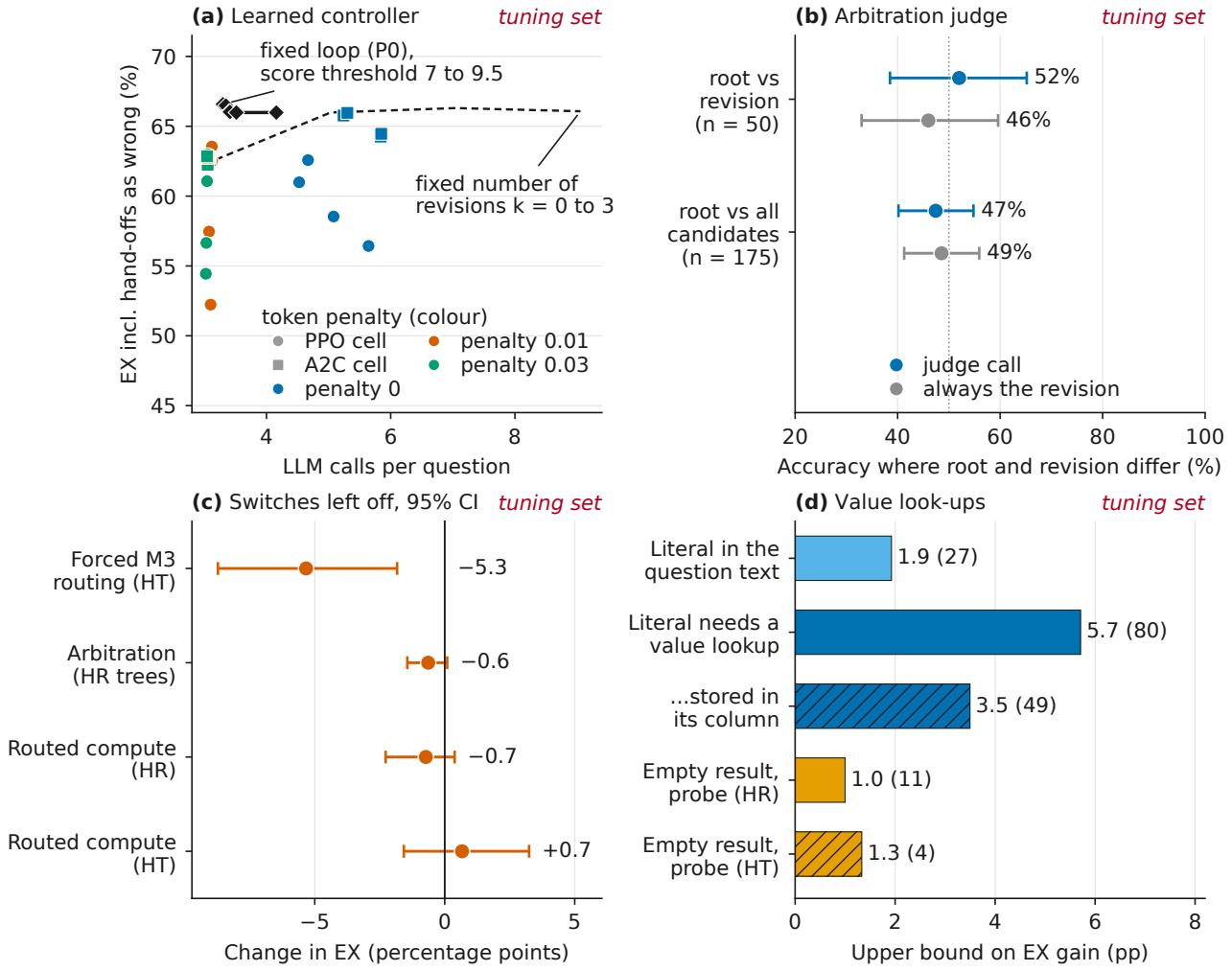
(a) Learned controller
tuning set
fixed loop (P0), score threshold 7 to 9.5
fixed number of revisions k = 0 to 3
token penalty (colour)
PPO cell
A2C cell
penalty 0
penalty 0.01
penalty 0.03
EX incl. hand-offs as wrong (%)
LLM calls per question
(b) Arbitration judge
tuning set
root vs revision (n = 50)
52%
46%
root vs all candidates (n = 175)
47%
49%
judge call
always the revision
Accuracy where root and revision differ (%)
(c) Switches left off, 95% CI
tuning set
Forced M3 routing (HT)
−5.3
Arbitration (HR trees)
−0.6
Routed compute (HR)
−0.7
Routed compute (HT)
+0.7
Change in EX (percentage points)
(d) Value look-ups
tuning set
Literal in the question text
1.9 (27)
Literal needs a value lookup
5.7 (80)
...stored in its column
3.5 (49)
Empty result, probe (HR)
1.0 (11)
Empty result, probe (HT)
1.3 (4)
Upper bound on EX gain (pp)